
When Accurate Forecasts Are Not Enough: A Physics-Structured Thermal World Model

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Abstract

1 Developing predictive feedforward at an operating 600 MW once-through boiler
2 exposed a practical gap: accurate steam-temperature forecasts need not describe
3 the effect of changing a spray valve. We construct a thermal world model in
4 which physical dynamics carry the action pathway, while a history-based observer
5 and bounded neural heat corrections address incomplete states and dynamics. On
6 shared internal-validation windows, a three-seed comparison gives main-steam
7 MAE of 0.371°C for a black-box forecaster, 0.969°C for a grey-box model, and
8 0.443°C for the GRU hybrid over 180 s. The hybrid’s prediction cost relative
9 to the black box is 0.0725°C (19.6%); its two valve responses are negative and
10 appreciable, whereas the black box is nearly insensitive in the registered diagnos-
11 tic. A tokenized cross-attention observer fails to improve MAE in all three seeds.
12 These results quantify prediction cost alongside model responsiveness, not plant-
13 response fidelity or demonstrated control benefit.

14 1 From temperature forecasting to valve evaluation

15 Our starting point was predictive feedforward development at a 600 MW coal-fired once-through
16 boiler that had operated for 12 years. Predictors could fit logged temperatures closely yet respond
17 weakly, or in an unexpected direction, to a changed spray-valve input. This author-reported mo-
18 tivation is not a quantified online intervention experiment. Anticipating excursions can support
19 warnings or disturbance feedforward; comparing valve corrections additionally requires a model of
20 their consequences.

21 In closed-loop logs, valves respond to temperatures and conditions; a predictor can exploit these
22 correlations without an independently informative action pathway Forssell and Ljung [1999].
23 Controlled-world-model theory links transition identifiability to conditional action excitation under
24 restrictive assumptions Zhang et al. [2026a]. This motivates our study mechanistically; its linear-
25 Gaussian bounds are not a theorem about the nonlinear boiler.

26 Physical modelling helps, but incomplete boundaries, latent thermal storage and unknown dis-
27 turbances complicate dynamic regimes. Fan-type models already combine balances with data-
28 identified terms Fan et al. [2020]; we do not claim their general failure. Accurate steam-temperature
29 forecasting Zhang et al. [2026b] does not settle action fidelity either. Following hybrid dynam-
30 ics Rackauckas et al. [2020], Yin et al. [2021], we ask: *can learning improve a physical thermal*
31 *model while keeping the valve pathway explicit, and what prediction cost accompanies that choice?*

32 We place learning in slow-state initialization and internal heat corrections, using one physical transi-
33 tion for forecasts and alternative-action rollouts. Our comparison reports main-steam accuracy, two
34 valve-response curves and a predeclared observer selection. This is an industrial method study, not
35 foundation pretraining, cross-plant transfer or controller deployment.

2 Learning inside the physical action pathway

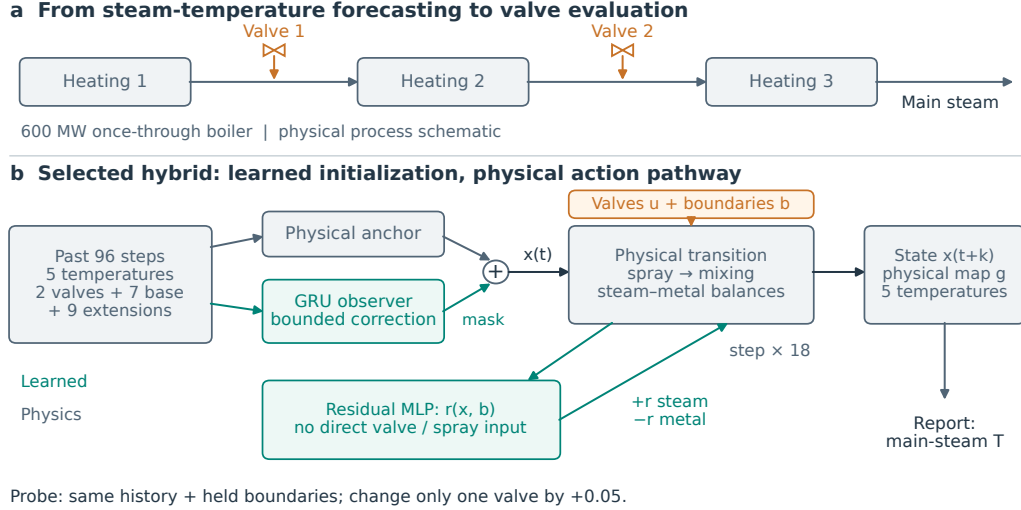


Figure 1: **The physical transition carries the action; neural modules complete the model.** The selected hybrid uses a past-only GRU and a bounded power closure (green). Valve-driven spray, transport, mixing and steam–metal balances define the grey-box rollout. The closure has no direct valve or measured spray-flow input, but depends on action-dependent states. Forecast and probe share weights and initialization. The diagram shows the mean path; rewetting is ablated in this hybrid.

State and inputs. The history $\mathcal{H}_t$ contains 96 samples at 10 s intervals: five temperatures, two valves, seven base boundaries and nine history-only extensions (23 variables). The 11-dimensional physical state contains three steam enthalpies, three metal temperatures, fuel lag, two liquid inventories and two spray-transport states. Valve positions determine spray flows, which affect downstream temperatures through transport and enthalpy mixing (Figure 1). A thermodynamic observation map produces five temperatures; our reported target is main steam.

Anchored initialization and heat closure. Five observed temperatures cannot specify every physical state. Their observation relations define a steady anchor x_{anc} ; a GRU supplies bounded corrections:

$$x_t = x_{\text{anc}}(\mathcal{H}_t) + M[0.1s_x \odot \tanh d_\phi(\mathcal{H}_t)], \quad (1)$$

$$r_k = R \tanh f_\theta(x_k, b_{k+1}^{\text{allow}}), \quad x_{k+1} = \Phi_\psi(x_k, u_{k+1}, b_{k+1}; r_k), \quad (2)$$

$$\hat{y}_{k+1} = g_\psi(x_{k+1}, b_{k+1}). \quad (3)$$

Here s_x denotes fixed state scales; M selects metal temperatures and fuel lag, leaving fast states anchored. The closure adds r_j to steam and $-r_j$ to metal power in each segment, with $R = 30,000$ kW. These are internal corrections, not an additive temperature-prediction head. Trainable physical parameters and 2 s semi-implicit substeps complete the transition. Rewetting gains are pinned near zero in both hybrid candidates; the grey-box reference retains rewetting.

Shared transition, bounded claim. At fixed history and boundaries, the diagnostic is $\widehat{\Delta T}_{1:H} = \hat{T}(\mathcal{H}_t, u^+, b) - \hat{T}(\mathcal{H}_t, u^0, b)$. Identical actions must yield identical outputs. Excluding direct valve and spray-flow inputs from f_θ prevents that direct shortcut, not state-mediated action dependence. Opposite power injections likewise do not guarantee correct response magnitudes. A token candidate replaces only the GRU observer with variable–time patches and state-query cross-attention; the physical rollout and closure remain unchanged (Appendix A).

3 A focused, frozen comparison

Data and common evaluation. One process side supplies 707,709 samples (about 82 days), split chronologically 75/15/10. We use internal validation only; the final test segment remains locked. Protocol revision 0.2 was frozen before this round of training, but earlier validation experiments informed its design. All four arms and seeds 0–2 use the same sampled 20,000-window training bank, 256 prediction windows and 64 response histories. The two evaluation sets each span 13 UTC days. Eligibility is fixed by valid base channels, finite in-range historical extensions and exact 10 s continuity. No results-based window filtering is applied.

Prediction supplies the recorded future valves and seven base boundaries over 18 steps. This is an *oracle-conditioned* comparison, not prospective forecasting with unknown future boundaries. The nine extensions, including water–coal ratio and load, are strictly history-only. Their normalization uses training rows only. The prediction metric is cumulative main-steam MAE,

$$E_H = \frac{1}{3} \sum_{s=0}^2 \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \frac{1}{n_d H} \sum_{i \in d} \sum_{h=1}^H |\hat{T}_{i,h}^{(s)} - T_{i,h}|. \quad (4)$$

Thus days, then seeds, receive equal weight; H18 averages 10–180 s, rather than measuring only the endpoint. Table 1 gives seed variability.

Method	H1	H6	H18
Persistence (shared reference)	0.085	0.268	0.639
Black box (variable tokens)	0.100 ± 0.013	0.191 ± 0.011	0.371 ± 0.007
Grey box (steady, no closure)	0.224 ± 0.003	0.529 ± 0.007	0.969 ± 0.008
Hybrid GRU (selected)	0.093 ± 0.007	0.224 ± 0.014	0.443 ± 0.022
Hybrid token (candidate)	0.111 ± 0.004	0.306 ± 0.019	0.654 ± 0.048

Table 1: **Main-steam prediction under the v0.2 information contract.** Cumulative MAE in °C, equal-day mean then three-seed mean ± sample standard deviation; persistence is shared and deterministic. H1/H6/H18 correspond to 10/60/180 s. These are validation-selected, fixed-budget results, not locked-test estimates.

Methods, not a fully matched architecture ablation. The black box is an iTransformer-inspired variable-token forecaster Liu et al. [2024], adapted to accept future covariates; it is not a reproduction of the published steam-temperature model. It and both observers receive the same 23 historical variables. The grey box has no learned observer or closure and consumes only the physical subset. The black box also consumes future measured total spray, which the action-driven physical transition ignores. Structured models train with five-temperature Gaussian NLL (24,000-update cap, batch 32); the black box uses main-steam MSE (3,000 updates, batch 128). All 12 runs reach their caps; reported checkpoints minimize validation H18 MAE. Information, loss, budget and rewetting differences preclude attributing cross-family differences to one architectural component.

Selection and response protocol. The token observer advances only if it improves H18 MAE in at least two of three paired seeds, has no higher seed-mean MAE and satisfies implementation gates. Valve responses are not selectors. For each probe, both valves and boundaries are held at their last historical values; one valve is increased by 0.05 of full opening, capped at 1.0. All realized doses are 0.05 within floating-point precision. Responses are equally averaged by day, then seed. All 64 histories pass the registered history-range-plus-0.05 support check for both valves; this marginal range check does not establish conditional excitation or joint counterfactual support. No field intervention truth is available.

88 4 Results: prediction cost and model responsiveness

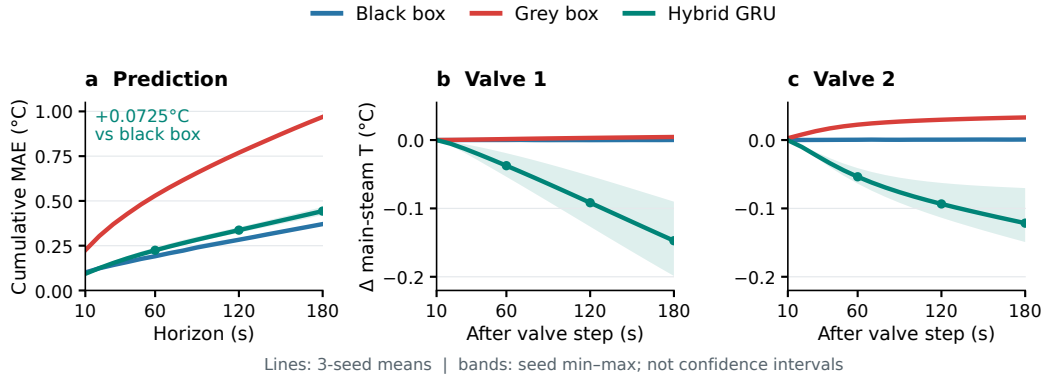


Figure 2: **The accuracy ranking does not describe the valve-response curves.** (a) Cumulative main-steam MAE on 256 common prediction windows. (b,c) Step-minus-base main-steam response on 64 common histories, fixed boundaries, one valve increased by 0.05. Lines average days then three seeds; bands show seed min-max, not confidence intervals. Both response panels share the same scale. All registered windows are retained. The responses are model outputs, not observed plant gains.

89 **A measurable forecast cost, not an accuracy win.** The GRU hybrid’s H18 MAE is 0.443°C , ver-
90 sus 0.371°C for the black box: $+0.0725^{\circ}\text{C}$ (19.6%). It is 54.3% lower than the grey-box reference’s
91 0.969°C . The ordering is horizon-dependent: GRU has slightly lower mean H1 error, but higher H6
92 and H18 error (Table 1). Paired day-block intervals put the hybrid-black-box H18 difference above
93 zero in two seeds; the third includes zero. These intervals are descriptive after validation selection,
94 not selection-adjusted tests (Appendix B).

95 **The best forecaster is nearly insensitive in this diagnostic.** At 180 s the black box’s seed-mean
96 responses are approximately $+0.00010$ and $+0.00073^{\circ}\text{C}$ for valves 1 and 2. The GRU hybrid in-
97 stead produces -0.147 and -0.122°C , with delayed cooling curves (Figure 2). Its H18 responses
98 are negative in all seeds for both valves, and all six endpoint day-block intervals exclude zero. The
99 grey box produces $+0.00448$ and $+0.03305^{\circ}\text{C}$: a physical state representation alone does not en-
100 sure the expected cooling direction in these fitted models. Because grey-box and hybrid rewetting
101 configurations differ, this is not an isolated proof that the neural components repair the sign. Nor do
102 near-zero black-box responses prove an error relative to an unmeasured plant gain.

103 **A more elaborate observer does not improve the compromise.** Token H18 MAE is 0.654°C ,
104 47.7% above GRU; it is worse in every seed and fails the predeclared advancement rule; its mean
105 also exceeds persistence (0.639°C). Its negative valve responses do not override that outcome. We
106 therefore retain GRU. This is a bounded negative result for this patch/query observer and budget,
107 not evidence against Transformers generally (Appendix C).

108 5 Discussion and conclusion

109 Our model combines learned thermal-state and heat corrections with an explicit, shared physical
110 action pathway. The present evidence locates a candidate compromise— 0.0725°C higher predic-
111 tion error than the black box alongside non-negligible cooling responses—not a calibrated optimum
112 or a Pareto frontier. Oracle inputs and reused validation limit generalization claims; differences
113 between model families limit causal architectural attribution. A matched no-rewetting grey-box con-
114 trol is now separately registered but pending; independent test and plant-response calibration remain
115 outstanding. For temporal world models intended to evaluate actions, forecast error and action re-
116 sponse should be reported together; neither the forecasting leaderboard nor physical structure alone
117 establishes decision readiness.

References

- He Fan, Zhi-gang Su, Pei-hong Wang, and Kwang Y. Lee. A dynamic mathematical model for once-through boiler-turbine units with superheated steam temperature. *Applied Thermal Engineering*, 170:114912, 2020. doi: 10.1016/j.applthermaleng.2020.114912.
- Urban Forssell and Lennart Ljung. Closed-loop identification revisited. *Automatica*, 35(7):1215–1241, 1999. doi: 10.1016/S0005-1098(99)00022-9.
- Yong Liu, Tengge Hu, Haoran Zhang, Haixu Wu, Shiyu Wang, Lintao Ma, and Mingsheng Long. iTransformer: Inverted transformers are effective for time series forecasting. *arXiv preprint arXiv:2310.06625*, 2024. doi: 10.48550/arXiv.2310.06625. Version 4.
- Christopher Rackauckas, Yingbo Ma, Julius Martensen, Collin Warner, Kirill Zubov, Rohit Supekar, Dominic Skinner, Ali Ramadhan, and Alan Edelman. Universal differential equations for scientific machine learning. *arXiv preprint arXiv:2001.04385*, 2020. doi: 10.48550/arXiv.2001.04385. Version 4, 2021.
- Yuan Yin, Vincent Le Guen, Jérémie Dona, Emmanuel de Bézenac, Ibrahim Ayed, Nicolas Thome, and Patrick Gallinari. Augmenting physical models with deep networks for complex dynamics forecasting. In *International Conference on Learning Representations*, 2021. URL <https://openreview.net/forum?id=kmG8vRXTFv>.
- Xiangteng Zhang, Yang Guan, Bo Zhang, Hongyang Li, Ya-Qin Zhang, and Shengbo Eben Li. On the identifiability of controlled world models. *arXiv preprint arXiv:2607.22430*, 2026a. doi: 10.48550/arXiv.2607.22430. Version 2.
- Yiyao Zhang, Feng Xie, Wei Shen, Xingyang Li, and Chase Wu. Prediction of superheated steam temperature in thermal power plants based on the iTransformer model. *Sensors*, 26(10):3078, 2026b. doi: 10.3390/s26103078.

141 A Implementation and information access

142 **Physical model.** The packed state is $(h_1, h_2, h_3, T_{m1}, T_{m2}, T_{m3}, r_b, m_{\ell 1}, m_{\ell 2}, d_{sw1}, d_{sw2})$.
 143 Temperature–enthalpy relations use a tabulated IAPWS property grid. The observation map reads
 144 physical states and supplied boundaries; there is no separate neural mean-temperature head. The
 145 steady anchor uses physical observation inversion. The hybrid corrects only metal temperatures and
 146 fuel lag; all three steam enthalpies and both liquid/transport pairs retain their anchored initial values.
 147 Mean corrections are bounded by 0.1 times fixed state scales. Zero-initialized observer heads return
 148 the anchor exactly. Five 2 s substeps form one 10 s sample step. The closure is recomputed at the
 149 current state of each substep; Equation 2 abbreviates this composition.

150 **Closure restrictions.** A two-hidden-layer tanh MLP returns three bounded power corrections, one
 151 per segment. The conservative mode applies equal and opposite steam/metal injections. Inputs are
 152 normalized physical state and six allowed base boundaries, excluding measured total spray. There
 153 are no additional latent dimensions or stochastic closure inputs in this round. Input exclusion is a
 154 direct-interface property: states already depend on earlier actions. With $S_k = \partial x_k / \partial \epsilon$ under an
 155 action perturbation, the local chain rule includes

$$S_{k+1} = (\Phi_x + \Phi_r r_x) S_k + \Phi_u \frac{\partial u_{k+1}}{\partial \epsilon}, \quad S_t = 0.$$

156 Thus the learned closure can still modify sensitivities through the state. The opposite injections
 157 constrain where residual heat enters, but do not prove parameter identifiability, numerical stability
 158 in every regime, or true gain recovery.

159 **Observer candidate.** Both hybrid observers have hidden width 128. The candidate patches each
 160 of 23 historical variables with patch length 16 and stride 8 (11 patches per variable, 253 tokens).
 161 Linear patch embeddings plus variable and patch embeddings enter one four-head Transformer en-
 162 coder; 11 learned state queries cross-attend to its output. Per-state correction heads receive the query
 163 features and pressure features. The GRU and candidate share the correction bound, state mask, phys-
 164 ical transition, closure, historical extensions and seed-specific training draws. The token candidate
 165 uses no future-history tokens. This is an observer replacement, not an end-to-end replacement of
 166 physical dynamics by attention.

167 **Black box.** Inspired by variable-token forecasting Liu et al. [2024], a linear 96-to-64 projection
 168 embeds each of 23 historical channels. A separate 18-to-64 projection embeds each of nine supplied
 169 future channels (two valves, seven base boundaries). Two four-head Transformer layers process the
 170 32 tokens; a flattened linear head predicts 18 main-steam deviations from the historical main-steam
 171 mean. Dropout is 0.1 and feedforward width is 128. This local adapted baseline should not be
 172 confused with a reproduction of the original benchmark architecture or the physics-guided steam-
 173 temperature predictor Zhang et al. [2026b].

Property	Black box	Grey box	Hybrid GRU	Hybrid token
Historical extensions consumed	9	0	9	9
Learned observer	—	—	GRU	Patch/query
Neural power closure	—	None	Conservative	Conservative
Rewetting	—	Enabled	Ablated	Ablated
Parameters receiving gradients	111,250	99	65,808	278,248
Update cap	3,000	24,000	24,000	24,000
Batch size	128	32	32	32
Validation frequency (updates)	500	200	200	200
Patience (evaluations)	3	20	20	20

Table 2: Executed configuration, not a capacity- or compute-matched benchmark. Active counts ex-
 clude instantiated but bypassed networks. In structured models, observation uncertainty parameters
 can receive gradients through NLL.

B Data contract, estimator and provenance

The seven base channels are steam flow, coal command, separator pressure, separator temperature, feedwater temperature, outlet pressure and total spray flow. The nine extensions are corrected fuel, active mill count, weighted mill gas temperature, flue oxygen, total secondary air, reheater gas inlet temperatures A/B, water-coal ratio and unit load. They inform history only: no recorded future extension is passed into any arm. Water-coal ratio is a covariate, not an additional physical power correction. Its isolated benefit is not identified without a feature ablation.

The physical model uses fixed engineering scales for base channels. Extensions use shared training-only means and standard deviations. The black box uses training-only statistics for all historical channels and the historical target mean for its output offset; its supplied future base covariates are not normalized by that historical transform. Future total spray is visible to the black box but ignored by action-driven physics. Holding it fixed during a valve change defines the diagnostic input, not a self-consistent plant counterfactual. These details are reasons to interpret the experiment as a comparison of complete methods.

Training draws use a fixed 20,000-window bank sampled with replacement. Validation and response manifests use seeds 30,000 and 60,000; the training-bank seed is 15,000. Each training seed controls initialization and subsequent minibatch draws. All models share identical validation targets, timestamps and starts; all probes share starts, valve doses and support masks. The prediction set has 256 windows and the probe set 64 histories, each covering 13 UTC days. Overlapping windows are possible; windows are not treated as independent experimental units.

For each seed, paired H18 error differences are averaged within UTC day. The 13 daily values are resampled with replacement 1,000 times, RNG seed 11,000; the 2.5th/97.5th percentiles form the reported interval. The same procedure applies pointwise to response curves. These intervals quantify day variation conditional on the selected checkpoints. They are not adjusted for checkpoint selection, previous validation reuse, multiple comparisons, or cross-day serial dependence. We do not treat three seeds as independent plants. Figure bands instead show the min-max range of three seed-level daily means.

H18 candidate minus reference (°C)	Seed 0	Seed 1	Seed 2
GRU – black box	0.068	0.046	0.104
95% day-block interval	[0.027, 0.108]	[−0.002, 0.093]	[0.067, 0.137]
GRU – grey box	−0.537	−0.541	−0.500
95% day-block interval	[−0.784, −0.318]	[−0.756, −0.341]	[−0.725, −0.299]
Token – GRU	0.267	0.212	0.154
95% day-block interval	[0.210, 0.329]	[0.155, 0.269]	[0.119, 0.193]

Table 3: Paired descriptive intervals, independently recalculated from saved arrays. No selection-adjusted or confirmatory significance claim is made.

The registered action support check expands each valve’s historical marginal minimum/maximum by 0.05 and checks both base and perturbed sequences. All 64 histories pass for each valve; no history was discarded. This permissive axis-aligned check does not test joint state-action support, conditional excitation or the realism of holding all supplied boundaries fixed. There is therefore no distinct supported-only curve in this round: it would be identical to the primary curve.

Artifact audit. The returned bundle contains 12 checkpoints, training ledgers and 24 raw prediction/response arrays. A local read-only audit verifies artifact hashes, checkpoint-to-input-identity binding and common index pairing. An independent saved-array reduction checks all source fingerprints, recalculates every prediction curve, response curve and paired interval, and recovers the same selected hybrid. Input, mapping and thermodynamic-grid fingerprints are retained in the internal audit record. The execution-side return reports 24 inference array replays; that statement was not independently rerun from the plant record in this manuscript-preparation pass. No training or test evaluation was performed to prepare this draft. The audit verifies the returned evidence, not the truth of an unmeasured plant intervention response.

215 C Observer selection and per-seed diagnostics

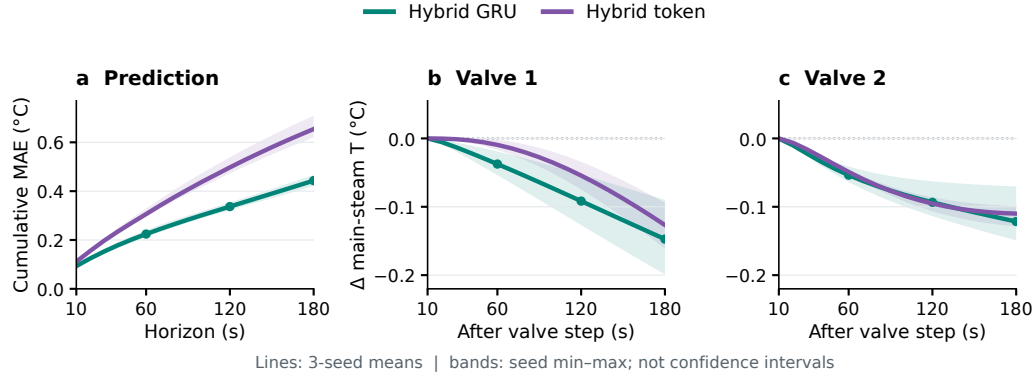


Figure 3: **The token observer does not advance.** Identical physical configuration, history contract and evaluation manifests; the observer architecture changes. Token prediction error exceeds GRU at H18 in all seeds, while both have negative mean endpoint responses. Responses are not model-selection criteria. Lines and bands use the same definitions as Figure 2.

Method / seed	H18 MAE	Valve 1 H18	Valve 2 H18	Best update
Black box / 0	0.3735	+0.000013	+0.001335	3,000
Black box / 1	0.3756	−0.000025	+0.000678	2,500
Black box / 2	0.3628	+0.000305	+0.000183	3,000
Grey box / 0	0.9787	+0.004601	+0.032581	23,800
Grey box / 1	0.9623	+0.004401	+0.033384	24,000
Grey box / 2	0.9668	+0.004435	+0.033175	24,000
Hybrid GRU / 0	0.4414	−0.198801	−0.145255	21,000
Hybrid GRU / 1	0.4216	−0.090115	−0.070247	23,600
Hybrid GRU / 2	0.4664	−0.152901	−0.149205	21,000
Hybrid token / 0	0.7086	−0.161024	−0.099171	23,800
Hybrid token / 1	0.6340	−0.094119	−0.128567	23,600
Hybrid token / 2	0.6203	−0.125057	−0.102259	22,000

Table 4: Seed-level daily means in °C; prediction is cumulative H18 MAE, response is the H18 endpoint difference, not the last-ten-step metric used in earlier development. All runs terminate at their caps.

216 For GRU, the valve-1 endpoint intervals for seeds 0/1/2 are respectively $[-0.229, -0.167]$,
217 $[-0.105, -0.075]$, $[-0.176, -0.128]$ °C; valve-2 intervals are $[-0.168, -0.119]$, $[-0.083, -0.057]$,
218 and $[-0.172, -0.123]$ °C. They support a negative model response under this probe. Small black-
219 box differences are close to the numerical granularity of large absolute-temperature predictions; we
220 neither equate them with exactly zero nor interpret a ratio to them as a physical amplification factor.

221 D Design history and remaining evidence boundaries

222 Earlier development investigated initial-state flexibility, residual placement, boundary prediction
223 and rewetting diagnostics. Those comparisons used different output aggregations or information
224 contracts; their MAEs are not pooled with the present main-steam-only metric. They motivated
225 the anchored hybrid and no-rewetting candidate, but are not confirmatory replications of this new
226 comparison. In particular, a prior five-temperature average and a current main-steam-only average
227 are not interchangeable, and earlier H60 probes do not extend the present registered 180 s evidence.

228 The remaining questions require different evidence. A one-time locked-test evaluation can address
229 temporal generalization after selection, but not true intervention gains. A matched grey-box no-
230 rewetting comparison would isolate the effect of adding the observer/closure more cleanly than the

231 present cross-family contrast. Independently calibrated intervention or suitably identified natural-
232 experiment responses would address plant-response magnitude. History-only prediction with fore-
233 cast rather than recorded future boundaries would address prospective use. After examining the
234 present results, the no-rewetting grey-box control has been separately registered with three seeds
235 and the original budget; it remains pending. Other checks remain proposals, not completed results
236 or silent changes to the existing protocol.

237 **Data availability and anonymization.** The data describe an operating industrial process. Public
238 release authorization for raw plant records has not been established; this draft makes no promise of
239 open raw data. Numerical figure sources, experiment configuration and audit records exist locally.
240 An anonymous sharing package and its permissions must be confirmed before submission; private
241 repository links and site identity are not included here.

242 **Use of language-model assistance.** Language-model tools assisted manuscript organization,
243 translation, figure-script development and consistency checks against saved results. They were not
244 used as a substitute for plant-response measurements. Human authors remain responsible for the
245 final scientific claims, numerical verification and submission.